AML Blind Spots (IS459 Group 2): Revised Questions and Pilot Findings

Why we changed: Pilot test revealed certain questions did not reveal meaningful kpis

Pilot scope: full IBM AML HI-Medium and LI-Medium sets, three files each (Trans.csv, Patterns.txt, accounts.csv)

- HI-Medium: 31,898,238 transactions, 35,230 laundering (1 in 905)
- LI-Medium: 31,251,483 transactions, 16,041 laundering (1 in 1,948)
- Pilot run locally on the complete files (DuckDB + Python), not on a sample, to test the logic before building on AWS.

Data checks that passed:

- Row and laundering counts match the published dataset exactly.
- 100% of transactions join to an account owner in accounts.csv (join on bank ID cast to integer + account number).
- 100% of Patterns.txt rows match a real transaction (HI 22,743 / 22,743; LI 3,909 / 3,909), so it is a usable answer key.

Question 1: Where does laundered money end up, how many hops does it take, and which accounts, entities and banks are the final cash-out points?

Who benefits and why

Financial-crime investigators and regulators. Money laundering is a chain of transfers, not a single transaction. Tracing money forward to where it stops shows which accounts, owners and banks are the real destinations. Investigators can then freeze those endpoints instead of chasing individual flagged transfers in the middle of a chain.

Pipeline

- Ingestion: upload HI and LI Trans.csv, Patterns.txt and accounts.csv to the raw S3 bucket with a boto3 Python script.
- Transformation:
    - AWS Glue (PySpark) cleans transactions to Parquet in a processed S3 bucket, partitioned by dataset and date, and registers them in the Glue Data Catalog.

    - The same job joins accounts.csv to add the owning entity and the bank's country.
    - A small parser job turns Patterns.txt into one row per known scheme: scheme ID, type, transactions.
- Computation:
    - A Glue job filters the 32M rows down to the laundering transactions (about 35K HI, 16K LI).
    - It runs the chain tracer (described below) and writes Q1 results to a results S3 bucket.
    - A second job scores the tracer against Patterns.txt.
- Presentation: Streamlit dashboard showing
    - a network graph of chains reaching a selected bank or entity
    - a hop-count histogram
    - a cash-out hub ranking
    - a validation page with accuracy by scheme type

How the tracer works

The tracer is a temporal BFS: a breadth-first trace that respects time.

- It processes laundering transactions once, in time order.
- A transfer continues a chain only if two conditions hold:
    1. the money arrived at the sender at or before the time of this transfer (money cannot move backwards in time)
    2. the amount is 0.8 to 1.25 times the amount received (the same money moving on, not an unrelated payment)
- The hop count is the fewest hops from the chain's origin, as in BFS.
- Because hops must move forward in time, chains cannot loop back on themselves.
- An account is a cash-out point if either:
    - it never passes the money on afterwards, or
    - the money leaves as cash, or as a cheque to a different owner.
- Accounts are ranked by how many separate chains end at them.

- **Keep only the laundering transactions** and line them up from oldest to newest.
- **Link each transfer to the money it passes on.** A transfer continues a chain if the sender received laundered money before sending, and the amount is roughly the same (within about 20%).
- **Count the hops.** Each time the money moves to a new account, the chain grows by one hop. If a transfer can't be linked to earlier money, it starts a new chain.
- **Find where the money stops.** An account is a cash-out point if the money never moves on from it, or leaves as cash or a cheque to someone else.
- **Summarise.** Rank the cash-out points by how many chains end there. Then group them by owner, bank and country.

Pilot findings

1. The tracer finds real endpoints and clearly beats simple rankings.

- It flags 96% (HI) and 97% (LI) of the true scheme endpoints in Patterns.txt as cash-out points.

- Share of the top 100 ranked accounts that are true endpoints:

Ranking	**HI**	**LI**
Tracer	100%	98%
Baseline: most laundered dollars received	37%	15%
Baseline: most laundering transfers received	89%	76%

- In LI the tracer's top 500 drops to 43%. The likely reason: Patterns.txt covers only 24% of LI laundering, so many real cash-out points have no answer-key entry.

2. Most laundering chains are short.

    - 98% (HI) and 99% (LI) of chains end within 2 hops.
    - Fewer than 1.2% (HI) and 0.3% (LI) of chains reach 5 or more hops. The longest is 13 hops.
    - Typical length by scheme type: about 1 hop for fan-in, fan-out and bipartite; about 2 for stack and scatter-gather; 4–5 for cycle and random schemes.

3. Hop counts are accurate in most cases.

    - Exact match with the answer key: 75% of HI schemes and 91% of LI schemes. Average error is 0.4 hops (HI) and 0.1 hops (LI).
    - Rules tested on the way:
        - the first version (longest path) was exact for only 56% of HI schemes
        - a plain temporal BFS reached 60%
        - adding the amount check gave the final 75%

4. Schemes share mule accounts.

    - 1,041 of 22,279 HI scheme accounts (4.7%) and 181 of 4,216 LI scheme accounts (4.3%) appear in more than one scheme.
    - This is why hop counts can never be 100% exact: when two schemes pass money through the same account, the money merges and no rule can fully separate it.

- For investigators, these shared accounts link separate schemes together and are high-priority targets.

5. Cash-out banks are consistent between the two datasets, but must be ranked by rate.

    - The bank ranking agrees between HI and LI (rank correlation 0.82 across 142 shared banks).
    - The top banks by raw count are simply the largest banks, each at about 20 cash-outs per 10K incoming transfers (close to the base rate). Final rankings will use laundering rate with a minimum volume.
    - By country, the US receives the most cash-outs (about 41% in HI), followed by China, Germany and France.

6. Owner-level view.

    - Grouping accounts by owner gives 17,280 cash-out owners in HI.
    - Among the top 100, 14 (HI) and 30 (LI) hold accounts at more than one bank. These are invisible to any single bank.

Caveats

- Patterns.txt covers only 64.6% of HI and 24.4% of LI laundering. Accuracy is measured on that subset.
- Rows dated Sep 17–28 are about 58% laundering. They are kept for Q1 because chains finish there.

---

2. Where does cross-border and currency risk sit in the transaction flow, and do standard red flags catch it?

- **2a. Cross-border corridors:** Are cross-border transfers more likely to be laundering than domestic ones, and which country-to-country corridors carry the highest laundering rate relative to their volume? Do official country-risk ratings (Basel AML Index) predict which corridors are risky?

- **2b. Currency pairs:** When money moves through an account, is it riskier when it arrives in one currency and leaves in another, and which currency pairs (received in X, sent in Y) carry the highest laundering rate? Would a standard rule that flags currency conversion within a single transaction catch it?

- **Who will benefit and why:** FX and cross-border payment compliance teams. Cross-border movement and currency changes can hide the trail of funds, and rules that check one transaction at a time may miss risk that only shows up across several transfers. The results tell these teams which corridors and currency pairs need tighter

screening, and whether common red flags (single-transaction conversion, official country-risk ratings) are worth the alerts they generate.

- **Ingestion:** Upload `HI-Medium_Trans.csv`, `LI-Medium_Trans.csv` and their `accounts.csv` files into the raw Amazon S3 bucket using a Python script with `boto3`. Upload the Basel AML Index country scores (external dataset) to the same bucket.

- **Transformation/Preparation:** AWS Glue (PySpark) cleans the transactions and joins `accounts.csv` to derive each bank's country (from the bank name). It adds a cross-border flag, a single-transaction conversion flag, a flag for currency changes between hops (the currency money arrived in vs the currency it leaves in), and the Basel AML Index score for each country. Output is written to S3 as date-partitioned Parquet and catalogued in the Glue Data Catalog.

- **Computation:** Amazon Athena runs SQL directly against the catalogued Parquet files to compute:

    - laundering rate and lift by corridor (with a minimum-volume threshold)
    - laundering rate by currency pair, both within a transaction and between hops
    - how many laundering transactions each standard red flag would catch, and how many alerts it would raise
    - laundering rate by Basel AML Index risk band
    - a comparison of the high-illicit (HI) and low-illicit (LI) datasets to check the results hold

- **Presentation:** The Streamlit dashboard connects to Athena and shows:

    - a country-to-country corridor heat map
    - a currency-pair heat map (received in X, sent in Y)
    - a red-flag scorecard comparing the alerts each rule raises with the laundering it catches
    - a Basel AML Index risk vs observed laundering chart

- **Preliminary pilot (proof of concept):** We ran the Q2 logic on the full HI-Medium and LI-Medium transaction files on a single laptop (DuckDB as a local stand-in for Glue and Athena), to confirm the signals exist before building the AWS pipeline.

    - Scope: Sep 1–16 only (later rows are mostly laundering and would inflate rates); transfers from an account to itself excluded. That leaves 29.3M transfers (HI) and 28.8M (LI).
    - Currency switches are judged only on what was known at the time of each transfer, as a bank would see it in real time.

Signal	HI laundering per 10K transfers	HI lift	HI share of laundering caught	LI laundering per 10K transfers	LI lift	LI share of laundering caught
All transfers (base rate)	10.6	–	100%	5.3	–	100%
2a. Domestic transfer	6.8	baseline	46%	4.3	baseline	57%
2a. Cross-border transfer	20.0	**2.9× domestic**	54%	7.8	**1.8× domestic**	43%
2b. Sender pays in a currency it has received before	6.8	baseline	54%	4.4	baseline	70%
2b. Sender pays in a currency it has never received before (currency switch)	208.5	**31× same currency**	28%	34.4	**7.8× same currency**	9%

- **Reading the table:** in HI, about 2 in every 100 currency-switch transfers are laundering, against 7 in every 10,000 same-currency transfers.
- **Two signals, two jobs:**
  - Cross-border is broad: it covers more than half of HI laundering but flags 8.4M transfers.
  - Currency switch is sharp: it flags only 415K transfers (about 1.4% of traffic) and is far more precise, but covers about a quarter of HI laundering.

- **Both signals hold in the low-illicit (LI) dataset,** just weaker. This is the robustness check the full pipeline will formalise.
- **Not yet done (full build):** corridor-level and currency-pair-level breakdowns, the red-flag scorecard and the Basel AML Index comparison.

X-factor

1. Proof the method works (Q1).

    - We score our chain tracing against IBM's answer key of 3,212 known laundering schemes (Patterns.txt), broken down by scheme type, on both a high-laundering (HI) and a low-laundering (LI) dataset.
    - Most analyses only describe results; ours shows they are correct.
    - Pilot: 98–100% of our top 100 cash-out accounts are true endpoints, against 15–37% for the dollar-amount baseline.

2. Linking accounts to their real owners (Q1).

    - accounts.csv maps accounts to their owners.
    - The cash-out hub becomes the controlling owner across banks, and we can show mule accounts shared between schemes (4.7% of scheme accounts in HI).

3. External country-risk data, used as a test (Q2).

    - We join the Basel AML Index (a continuous country risk score, richer than the yes/no FATF list) through the country in each bank's name, and test whether official ratings would have found the risky corridors.
    - Pilot: they would not (negative correlation).
    - This gives a business takeaway: corridor screening should be driven by observed transaction data, not by published ratings alone.

Tell Shutdown dev link use prod link from now on and share updates are made